PROJECT SPARK — LEVEL 22 CREATION GUIDE

One Way Only — Diode Direction | 5 V | 330 Ω | 1N4007 | Green LED

Create the ORIGINAL Level_22_Layout prefab; keep Level 21 unchanged. The supplied concept illustration is not a native Unity scene.

1. Create and import

• Back up the Unity project. Open the original Level_22_Layout prefab in Prefab Mode.

• Select the original Level_22.asset and import Level_22_Diode_Direction_Fixed.txt with your ScriptableObject text importer.

• Confirm levelId=level_22, requiredTargetCount=4, and source PowerSupply_01 at 5.0 V.

2. Place components

• Add PowerSupply_01 (5 V), Power_Switch, a healthy 330 Ω resistor, SparkDiode (1N4007), and a green SparkLED.

• Add Diode_Slot_01 and Register_Slot_LIMIT as SparkShelfSlot workstations. Install shelf items through the existing registration path.

• Diode cathode is the silver-banded end. Place it toward the LED anode. The green LED cathode faces the battery negative return.

3. Exact wiring (five external wires)

• W1: Battery_positive → Power_Switch INPUT

• W2: Power_Switch OUTPUT → R1 INPUT

• W3: R1 OUTPUT → D1 ANODE

• W4: D1 CATHODE (band) → LED1 ANODE

• W5: LED1 CATHODE → Battery_negative

• Do not put a direct wire across a diode, switch, resistor or LED, and do not bypass them.

4. Bind runtime IDs

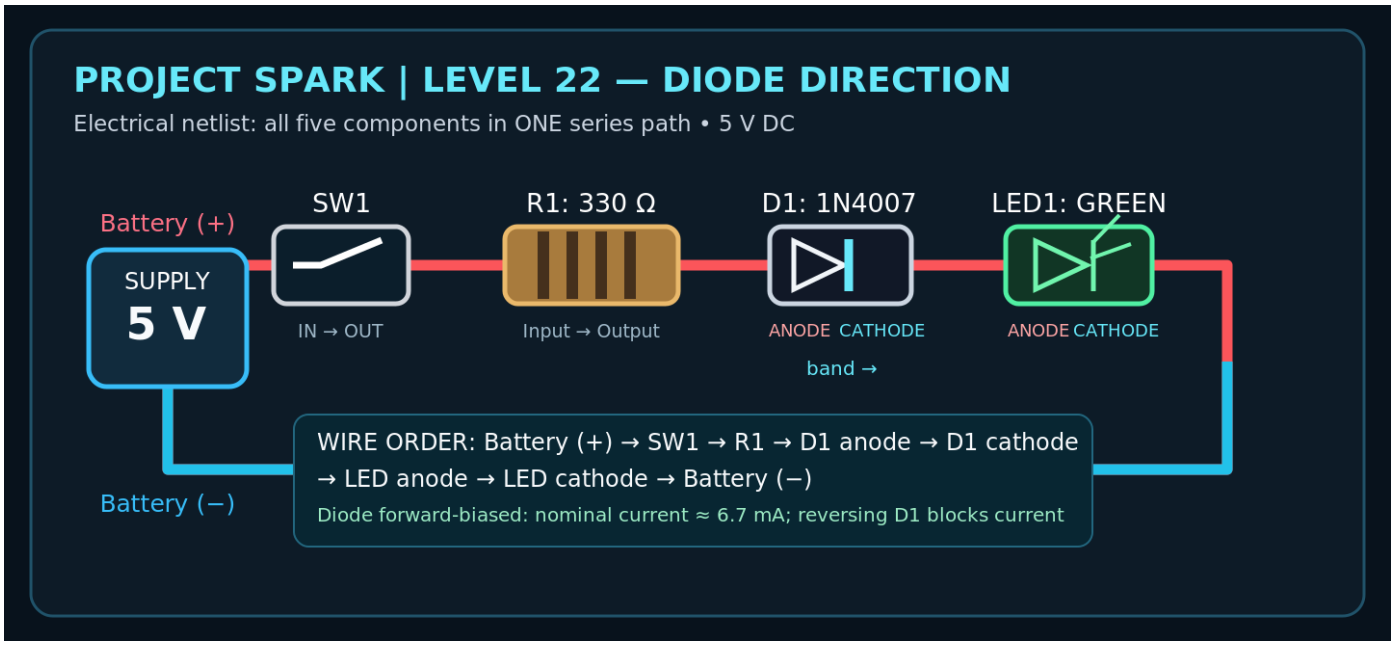
• Diode_Slot_01: D_PROTECT_Component / D_PROTECT_Anode / D_PROTECT_Cathode; electrical type SparkDiode.

• Register_Slot_LIMIT: R_LIMIT_Component / R_LIMIT_Input / R_LIMIT_Output; electrical type SparkResistor.

• Fixed green LED: LED_INDICATOR_Component / LED_INDICATOR_Anode / LED_INDICATOR_Cathode using SparkLevelSceneBindings.

• Use Smart Slots ONLY when each workstation has correct smart IDs; otherwise use explicit legacy mappings. Exact terminal names must match the actual GameObjects.

LEVEL 22 — EXACT CIRCUIT SCHEMATIC



5. Target checks

• level22_supply — 4.8–5.2 V across Battery_positive / Battery_negative.

• level22_diode_forward — 0.45–1.0 V across D_PROTECT_Anode / D_PROTECT_Cathode.

• level22_led_on — SparkLED must be ON; fixed LED binding must resolve.

• level22_safe_current — 0.005–0.008 A magnitude through R_LIMIT_Component.

• Nominal expectation: approximately 6.67 mA, diode drop 0.7 V, green LED drop 2.1 V; validate against actual solver results.

6. Play Mode and debug

• With switch OFF, verify wiring and inspect the runtime bridge; close switch and read real solved current and diode/LED states.

• Reverse the diode only with the supply OFF: forward orientation should light the green LED; reversed should block it.

• Run Synchronize + Report Now, then inspect [LEVEL TARGET RESULT]. All component/terminal bindings must be RESOLVED and 4/4 targets Passed=True.

• If diode is UNBOUND, check its shelf occupant, runtime declarations, bridge electrical type and exact terminal names; do not edit the diode solver to hide a binding error.

7. Whiteboard pages

• Page 1: polarity and band. Page 2: five-wire circuit. Page 3: forward vs reverse LED test. Page 4: current, voltage and four targets.

Package note: The illustrated workbench is a visual reference only, not a verified Unity export.